MIL-N-3497C
1 November 1978
SUPERSEDING
MIL-N-3497B
29 July 1970

MILITARY SPECIFICATION

NADIRSCOPE, STEREOPLOTTER TRACING TABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a nadirscope attachment for the multiplex tracing table used in photogrammetric mapping.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

C-F-206	- Felt Sheet: Cloth, Felt, Wool, Pressed.
QQ-B-613	- Brass, Leaded and Nonleaded: Flat Products (Plate, Bar, Sheet, and Strip).
TT-V-121	- Varnish, Spar, Water-Resisting.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

FSC 6675

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-116	- Preservation-Packaging, Methods of.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-F-2312	- Felt, Hair or Wool: Mildew Resistant, and Moisture Resistant Treatment for.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

DRAWINGS

ME

TA13217E7491	- Nadirscope, Stereoplotter Tracing Table.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The nadirscope shall be in accordance with Top Assembly TA13217E7491 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the nadirscope.

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Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (first produced nadirscope). The contractor shall furnish one nadirscope for examination and testing within the time frame specified (see 6.2) to prove that his production methods will produce nadirscopes that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of the specification (see 6.5).

3.4 Environmental conditions. The nadirscope and case shall not be damaged by storage in any ambient temperature from plus 155° F (plus 68.3° C) for a period of four hours to minus 50° F (minus 45.6° C) for a period of six hours with no solar radiation.

3.5 Case. The case shall be hardwood with hinged cover and latching fastener. The hinges and fastener shall be brass conforming to QQ-B-613, Annealed, Composition 268. The case shall be finished with two coats of spar varnish conforming to TT-V-121.

3.5.1 Lining. The case shall be lined with wool felt conforming to C-F-206, Type I, Classification 16R2, 1/16-inch thick, Color Green. The wool felt shall be treated for mildew and moisture resistance in accordance with MIL-F-2312, Type III, with 2,2' methylene bis - (4-chlorophenol).

3.6 Identification marking. The nadirscope and case shall be identified in accordance with MIL-STD-130.

3.7 Treatment and painting. The portion of the nadirscope normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, Type A.

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3.8 Government-loaned property. Unless otherwise specified (see 6.2), the following property in the quantity indicated will be loaned by the Government (see 4.5.2.2 and 6.4).

Item No.	Description	Identification	Quantity
1.	Multiplex tracing table.	NSN 6675-00-371-9492	1

3.9 Workmanship. All parts and components of the nadirscope shall be free from burrs, sharp edges, and corners except when sharp edges and corners are required.

3.9.1 Machine work. Tolerances and gages for metal fits shall conform to the limitations specified herein and on the applicable drawings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Disassembly inspection. Failure of any examination or test by the first produced nadirscope shall be cause for disassembly, in the presence of a Government representative, of the first produced nadirscope to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the first produced nadirscope. Reassembly with replacement parts and retesting shall be the responsibility of the contractor.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First produced nadirscope inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 First produced nadirscope inspection.

4.3.1 Examination. The first produced nadirscope shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection or for performing the disassembly inspection specified in 4.1.1.

4.3.2 Tests. The first produced nadirscope shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection or for performing the disassembly inspection specified in 4.1.1.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.4.2 Examination.

4.4.2.1 Samples. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 1.0 percent defective for major defects and 6.5 percent defective for minor defects.

4.4.3 Tests.

4.4.3.1 Samples. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2. AQL shall be 1.0 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The nadirscope shall be examined as specified herein for the following defects:

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Major

101. Dimensions not as specified.
102. Crosshairs not sharp or accurately marked.
103. Material not as specified.
104. Case not as specified.
105. Lining not as specified.
106. Identification marking missing, incomplete, or illegible.
107. Workmanship not as specified.

Minor

201. Loose or cracked joints in case.

4.5.2 Tests.

4.5.2.1 Environmental.

4.5.2.1.1 Low temperature storage. The nadirscope and case shall be subjected to a temperature of minus 50° F (minus 45.6° C) for a period of six hours with no solar radiation. Upon completion of the low temperature test the instrument and case shall be examined.

4.5.2.1.2 High temperature. The nadirscope and case shall be subjected to a temperature of plus 155° F (plus 68.3° C) for a period of four hours. Upon completion of the high temperature test the instrument and case shall be examined.

4.5.2.1.3 Failure criteria. Any evidence of deformation of components, binding of the reticle mount and retainer, distortion of the reticle crosslines, peeling or chipping of paint or varnish, cracking or warping of the wooden case, or discoloring or bulging of the felt case lining shall constitute failure of the environmental tests.

4.5.2.2 Test conditions. Tests shall be performed with the nadirscope attached to a Government-loaned multiplex tracing table specified in 3.8.

4.5.2.3 Fit. The front and rear slide rod spacers shall be securely attached in using position to the slide rods. The slide rods shall then pass through the clamping yoke for their entire effective length.

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4.5.2.3.1 Failure criteria. Adverse binding or looseness of fit of the slide rods as they pass through the clamping yoke at any point shall constitute failure of this test.

4.5.2.4 Operation. The reticle carriage shall be locked in operating position by means of a knurled locking screw, and the crosslines shall be carefully centered over the dot of light in the center of the tracing-table platen. Gentle pressure shall be applied to the free end of the reticle carriage to assure that no play exists in any part of the instrument. The locking screw shall be loosened, and the reticle carriage shall be moved completely out of operating position. The reticle carriage shall be returned to the operating position and locked with the locking screw. This shall be repeated three times, and each time the crosslines on the reticle should return to a point exactly above the dot of light.

4.5.2.4.1 Failure criteria. Failure of the crosslines on the reticle to return exactly above the dot of light in center of tracing-table platen after moving the reticle carriage out of and back into operating position shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

108.	Materials, methods, and containers not as specified for Level A or Level B. Each incorrect material, method, or container shall be considered one defect.
109.	Nadirscope not contained in case as specified for Level A or Commercial.
110.	Contents exceed the weight or size limitations of the box for Level A or Level B.

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111. Box closure and strapping not in accordance with the referenced document for Level A or Level B.
112. Marking missing, illegible, incomplete, or incorrect for Level A, Level B, or Commercial.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial, as specified (see 6.2).

5.1.1 Level A. The nadirscope, contained in the case, shall be preserved in accordance with MIL-P-116, Method IC-1 or IC-3.

5.1.1.1 Consolidation. The nadirscope, preserved as specified in 5.1.1, shall be consolidated together in a close-fitting box conforming to PPP-B-636, W6c in quantities not to exceed the weight or size limitations of the box.

5.1.2 Commercial. The nadirscope, contained in the case, shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, Level B, or Commercial, as specified (see 6.2).

5.2.1 Level A. The nadirscope, preserved as specified in 5.1, shall be packed together in a close-fitting box conforming to PPP-B-601, Overseas Type, style optional. Box closure and strapping shall be in accordance with the appendix to the box specification.

5.2.2 Level B. The nadirscope, preserved as specified in 5.1, shall be packed together in a close-fitting box conforming to PPP-B-636, V3c, V11c, V13c, or V15c in quantities not to exceed the weight or size limitation of the box specification. Box closure and strapping shall be in accordance with the appendix to the box specification.

5.2.3 Commercial. The nadirscope, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection shall be in accordance with MIL-STD-129.

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5.3.2 Commercial. Marking for Commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The nadirscope is intended for use in photomapping as an accessory to the multiplex tracing table to determine the nadir point of multiplex projectors.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of first produced nadirscope (see 3.2).
(c) When Government-loaned property is not required (see 3.8).
(d) Degree of preservation and degree of packing required (see 5.1 and 5.2).

6.3 First produced nadirscope. Any changes or deviations of production nadirscopes from the approved first produced nadirscope during production will be subject to the approval of the contracting officer. Approval of the first produced nadirscope will not relieve the contractor of his obligation to furnish nadirscopes conforming to this specification.

6.4 Government-loaned property. The contracting officer should arrange to loan the property specified in 3.8, when applicable (see 6.2).

6.5 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

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DOCUMENT IDENTIFIER AND TITLE

MIL-N-3497C Nadirscope, Stereoplotter Tracing Table

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

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